Cell Division II

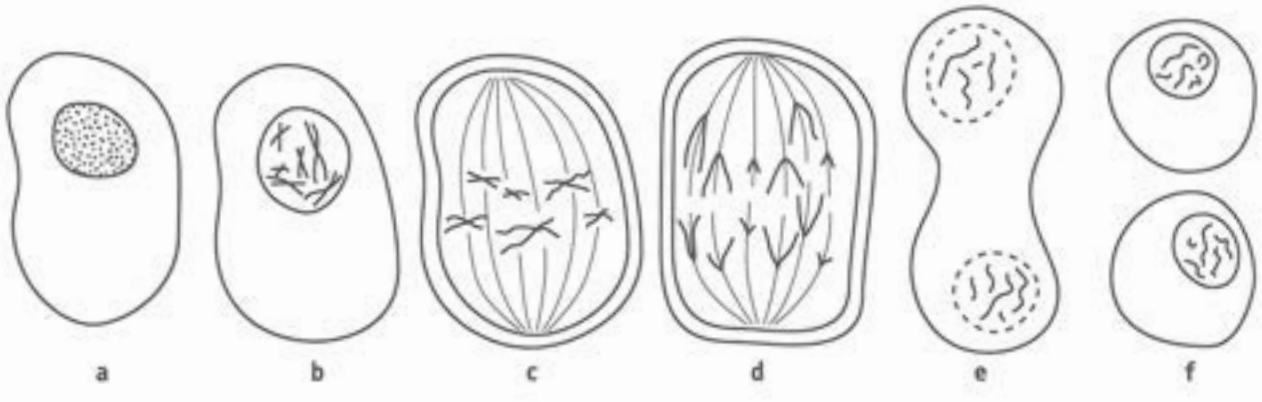
11A

When human sex cells (gametes) join at fertilisation they produce a single cell (a *zygote*). This cell starts to divide and, after nine months (and billions of cell divisions), a baby is born.

There are thin, thread-like, microscopic structures called chromosomes present in the nucleus of a cell. The chromosomes can only be seen when a cell is dividing. Chromosomes occur in cells in pairs (apart from in sex cells). Humans have 23 pairs ($2n$) of chromosomes. Other species have different numbers of chromosomes. Chromosomes are strands of DNA (deoxyribonucleic acid), and consist of sections called genes. These contain the instructions for controlling cells, and need to be able to be copied precisely, because a mistake made early in replication will be repeated in every cell.

There are two processes by which cells can divide: *mitosis* and *meiosis*. In mitosis the cell continues to duplicate and divide, producing daughter cells that are genetically identical to the parent cell. These are needed for growth and to replace the millions of cells that die every day.

1 Mitosis occurs in *somatic* cells. What are 'somatic' cells?



This diagram shows a simplified version of the sequence followed in mitosis (nucleus is enlarged for clarity).

a The parent cell is resting, no chromosomes are visible.

b Pairs of duplicated chromosomes, 'sister chromatids', become visible. (**prophase**)

c The sister chromatids line up along the centre ('equator') of the cell. (**metaphase**)

d The sister chromatids split and move to opposite poles. (**anaphase**)

e Two new daughter nuclei formed. Cell splits into two. (**telophase**)

f Each new (daughter) cell has the same number of chromosomes as the parent cell.

Mitosis occurs in most body cells. It also happens in organisms that reproduce asexually.

> Hint: Mitosis produces twice the numbers of chromosomes as meiosis. Remember this by **twice** and mi**tosis**. Or, remember that the first three letters of mitosis are '**mit**', the same as '**mit**tens', and there are two mittens in a pair.

2 What is meant by 'asexual reproduction'?

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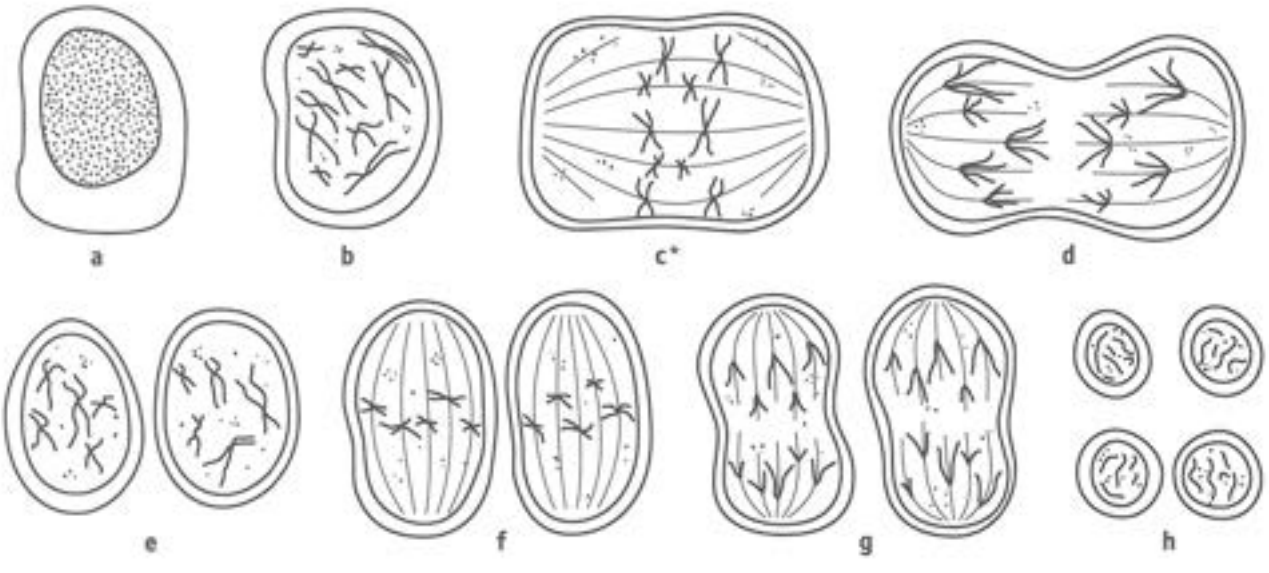
Cell Division II

11B

Meiosis occurs when sex (germ) cells are produced. If the sex cells in humans each contained the full number of chromosomes, then the zygote would have 46 pairs. The number of chromosomes present in a 'normal' body (somatic) cell is called the *diploid* number. Meiosis produces four daughter cells, each of these having half the number of chromosomes (n) of the original parent cell (that is the *haploid* number).

Meiosis (from the Greek, meaning 'to make smaller') is more complex than mitosis. Both processes involve four phases — prophase, metaphase, anaphase and telophase — but meiosis involves extra processes. For this reason, the phases (a) to (e) below are labelled prophase I, metaphase I and so on, and from (e) to (h) they are labelled prophase II, metaphase II, etc.

The diagram and steps listed below describe a simplified sequence for meiosis (nucleus enlarged).



a Parent cell (no chromosomes visible).
b Chromosomes become visible due to 'thickening'.
c Sister chromatid pairs pair up and align at 'equator'. (*Note: pairs of sister chromatids from *both* parents are present.)
d Chromatid pairs divide for the first time.
e Cell divides for the first time.
f Sister chromatids align for the second time.
g Second cell division.
h Four daughter cells formed, each with half the original number of chromosomes, all genetically different.

Meiosis helps cause genetic variation. One stage causes genes to mix. This helps ensure that new organisms are never identical to their parents. Humans can produce over 8 million types of sperm or ova, and there are 70 000 000 possible ways that these gametes can join.

3 Describe (with the aid of diagrams) what happens in the prophase.

4 Describe (with the aid of diagrams) what happens in the metaphase.

5 Describe (with the aid of diagrams) what happens in the anaphase.

6 Describe (with the aid of diagrams) what happens in the telophase.

7 Give one reason why meiosis only produces half (n) the chromosomes as mitosis ($2n$).

8 Assuming the population of the world is around 5 billion, why is it very unlikely to have two unrelated people with the same genetic make up?

9 Name two major differences between mitosis and meiosis.

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